

Materials & Filament Guide

Choose the Right Filament Every Time



3D LAMP PACK™

Filament Comparison

FILAMENT	LIGHT DIFFUSION	STRENGTH	PRICE / KG	BEST FOR
PLA	Good	Medium	\$15–\$20	Most lamps, beginners
PLA+ RECOMMENDED	Good	High	\$18–\$25	Best all-around choice
PETG	Excellent	High	\$20–\$28	Outdoor lamps, durability
TPU	Poor	Very High	\$25–\$35	Flexible accent pieces
Silk PLA	Poor	Medium	\$20–\$30	Decorative, premium look
ASA / ABS	Medium	High	\$18–\$25	Outdoor, UV-resistant

Color Recommendations

For Warm Light (2700–3000K LEDs)

- **White / Natural PLA** — Maximum diffusion, warm cozy glow. Best seller.
- **Light Yellow / Cream** — Enhances warm tones, feels like candlelight.
- **Light Orange** — Deep amber glow, great for bedroom and ambient lamps.

For Cool Light (4000–6500K LEDs)

- **White / Transparent** — Clean modern look, excellent for office and desk lamps.
- **Light Blue** — Futuristic, tech-inspired vibe.
- **Light Green** — Nature-themed, works well with plant-inspired designs.

Pro Tip: Silk PLA filaments (gold, silver, copper, rose) look stunning when the lamp is off — they have a shiny, metallic surface. But they block most light. Use them for opaque bases, decorative caps, or lamps designed to cast light upward or downward rather than through the walls.

How Much Filament Per Lamp?

LAMP SIZE	FILAMENT USED	COST (PLA)	PRINT TIME
Small (10–12 cm / desk lamp)	30–60 g	\$0.50–\$1.00	2–4 hrs
Medium (15–20 cm / table lamp)	60–120 g	\$1.00–\$2.00	4–8 hrs
Large (25–35 cm / pendant)	120–250 g	\$2.00–\$4.50	8–16 hrs

Electrical Components

USB LED Kits (Easiest Option)

USB-powered LED puck lights or fairy-light bundles are the simplest way to illuminate your lamps. They plug into any USB port or power bank, require no wiring, and cost \$2–\$5 each. Search Amazon for "USB LED lamp base" or "USB puck light warm white." Buy in bulk (10–20 pack) to bring the per-unit cost under \$2.

E27 / E26 Cord Sets (For Pendant Lamps)

For hanging pendant lamps, use a standard E26 (US) or E27 (EU/UK) cord set with an inline switch. These come with a ceiling canopy, 5–15 ft cord, and socket — ready to hang. Pair with a 5W–7W LED bulb. Cost: \$6–\$12 per cord set.

LED Strip Integration

For lamps with channels, grooves, or internal cavities, adhesive-backed LED strips provide even, continuous lighting. Use 2835 or 5050 SMD strips in warm white. Cut to length at the marked intervals. A 5-meter roll (\$8–\$15) is enough for 10–15 small lamps.

Safety First: PLA softens at ~60 °C. Always use LED lighting (not incandescent or halogen). Keep wattage at 5W or below for fully enclosed shades. Ensure all wiring connections are properly insulated. Never sell lamps with exposed wiring.

Where to Source Components

COMPONENT	SOURCE	UNIT COST	BULK COST (10+)
USB LED base / puck	Amazon	\$3–\$5	\$1.50–\$2.50
E26/E27 cord set	Amazon	\$8–\$12	\$5–\$8
5W LED bulb (warm)	Amazon	\$2–\$4	\$1–\$2
LED strip (5 m)	Amazon / AliExpress	\$8–\$15	\$5–\$10
Inline dimmer switch	Amazon	\$3–\$6	\$2–\$4
Touch dimmer base	AliExpress	\$4–\$8	\$2.50–\$5

Startup Budget: Your First 10 Lamps

10-Lamp Starter Budget

PLA+ Filament (1 kg — makes 10–15 small lamps)	\$20
USB LED bases × 10 (bulk)	\$20
Packaging (small boxes + tissue, 10 units)	\$15
Etsy listing fees (10 listings × \$0.20)	\$2
Electricity (~30 hrs printing)	\$3
Total Startup Investment	~\$60

Return on Investment: At \$35–\$50 per lamp, selling all 10 generates \$350–\$500 in revenue from a \$60 investment. That is a 5–8x return before scaling.

Next Step → Order a 1 kg spool of white PLA+ and a 10-pack of USB LED bases. Start printing your first batch today.